



deeplearning.ai

Object Detection

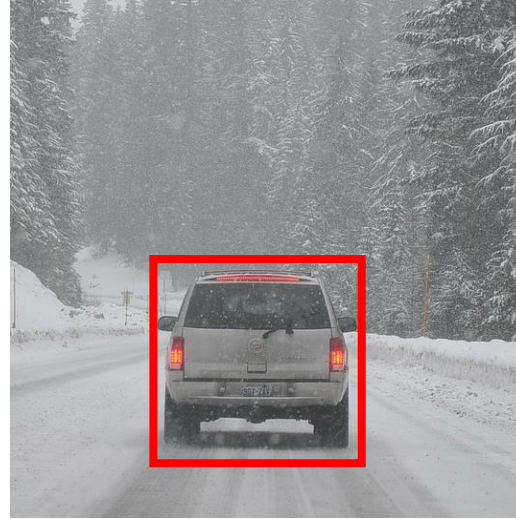
Object
localization

What are localization and detection?

Image classification



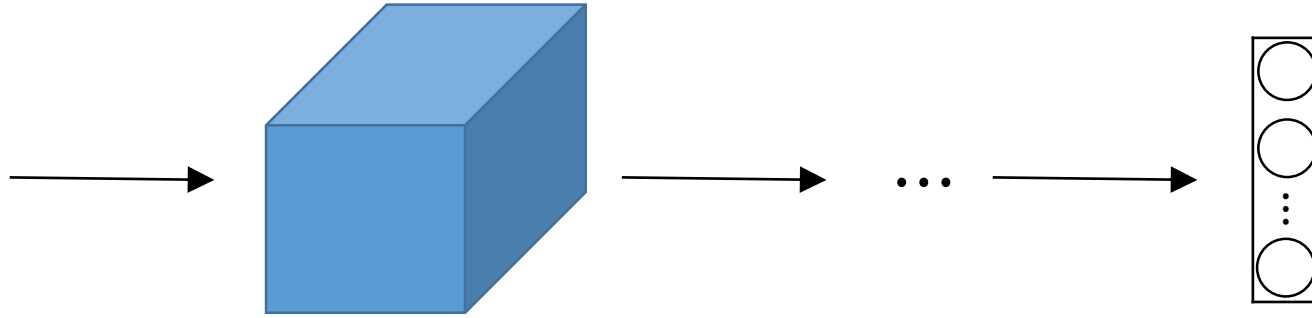
Classification with localization



Detection



Classification with localization

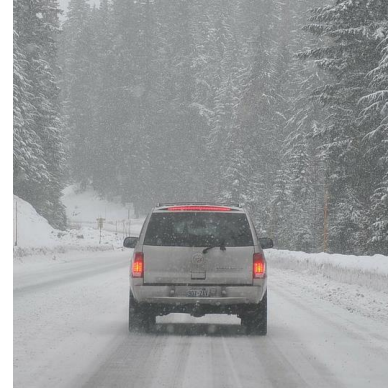


- 1 - pedestrian
- 2 - car
- 3 - motorcycle
- 4 - background

Defining the target label y

- 1 - pedestrian
- 2 - car
- 3 - motorcycle
- 4 - background

Need to output b_x, b_y, b_h, b_w , class label (1-4)



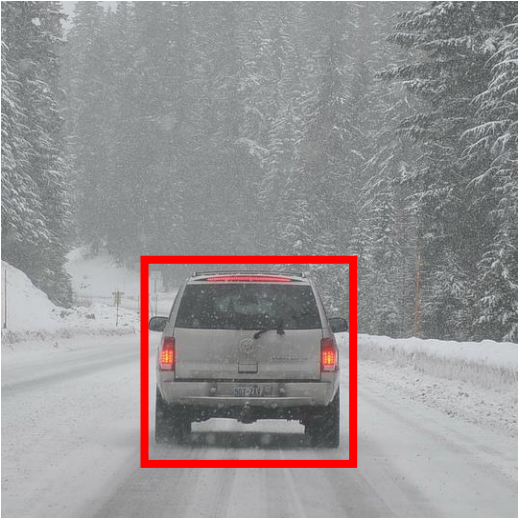
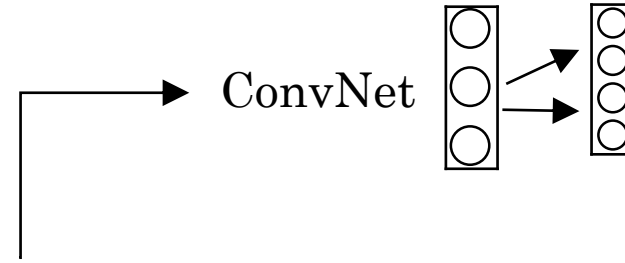


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Object Detection

Landmark
detection

Landmark detection



b_x, b_y, b_h, b_w





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Object Detection

Object
detection

Car detection example

Training set:

x

y



1



1



1



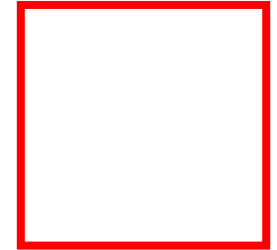
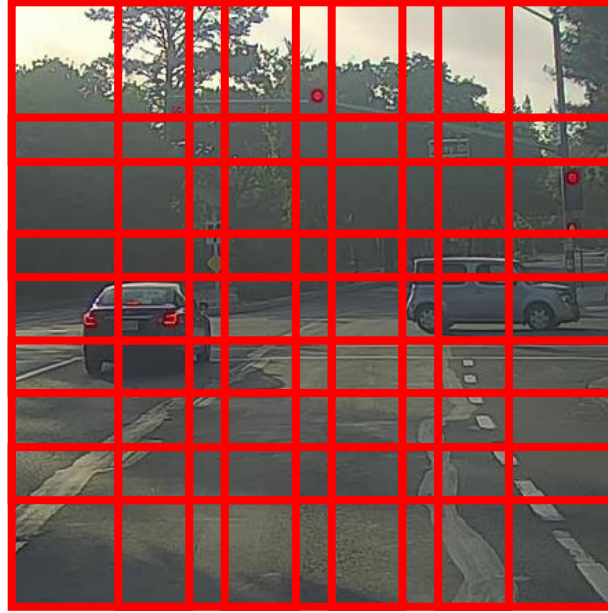
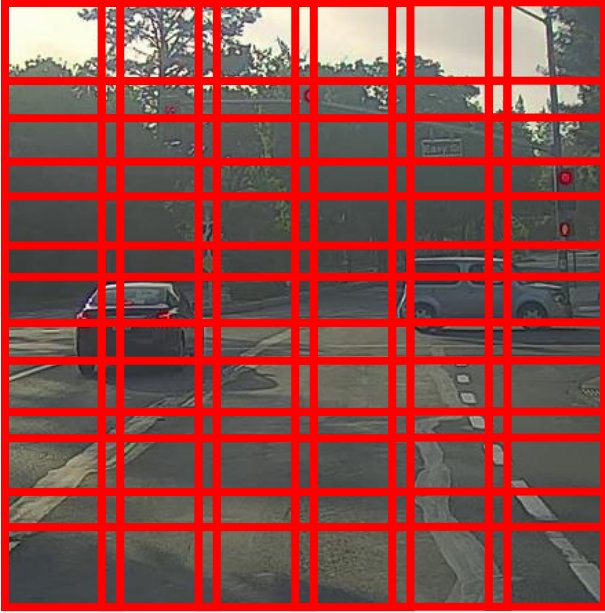
0



0



Sliding windows detection



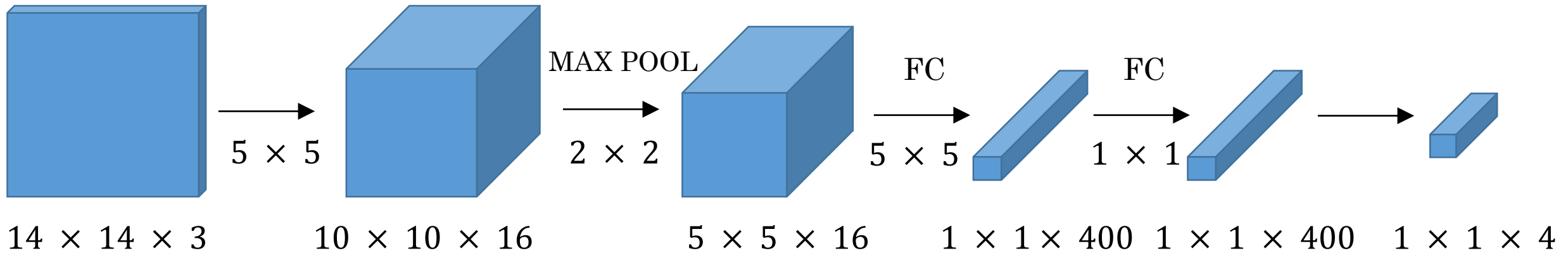
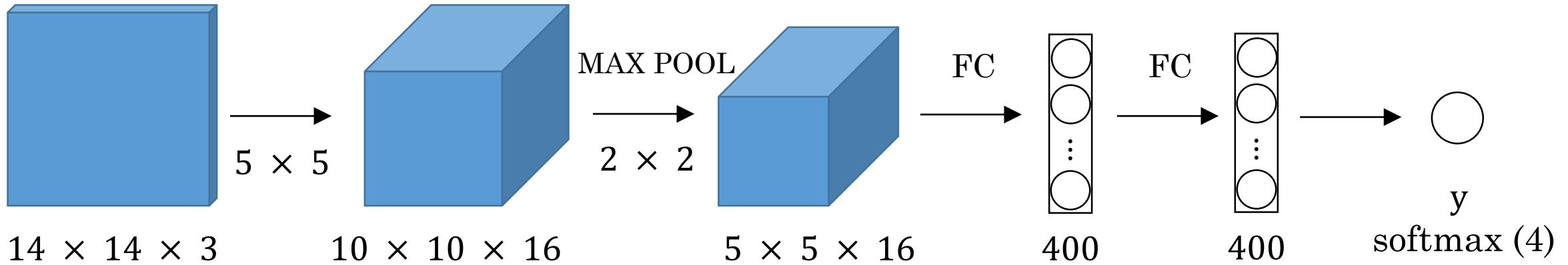


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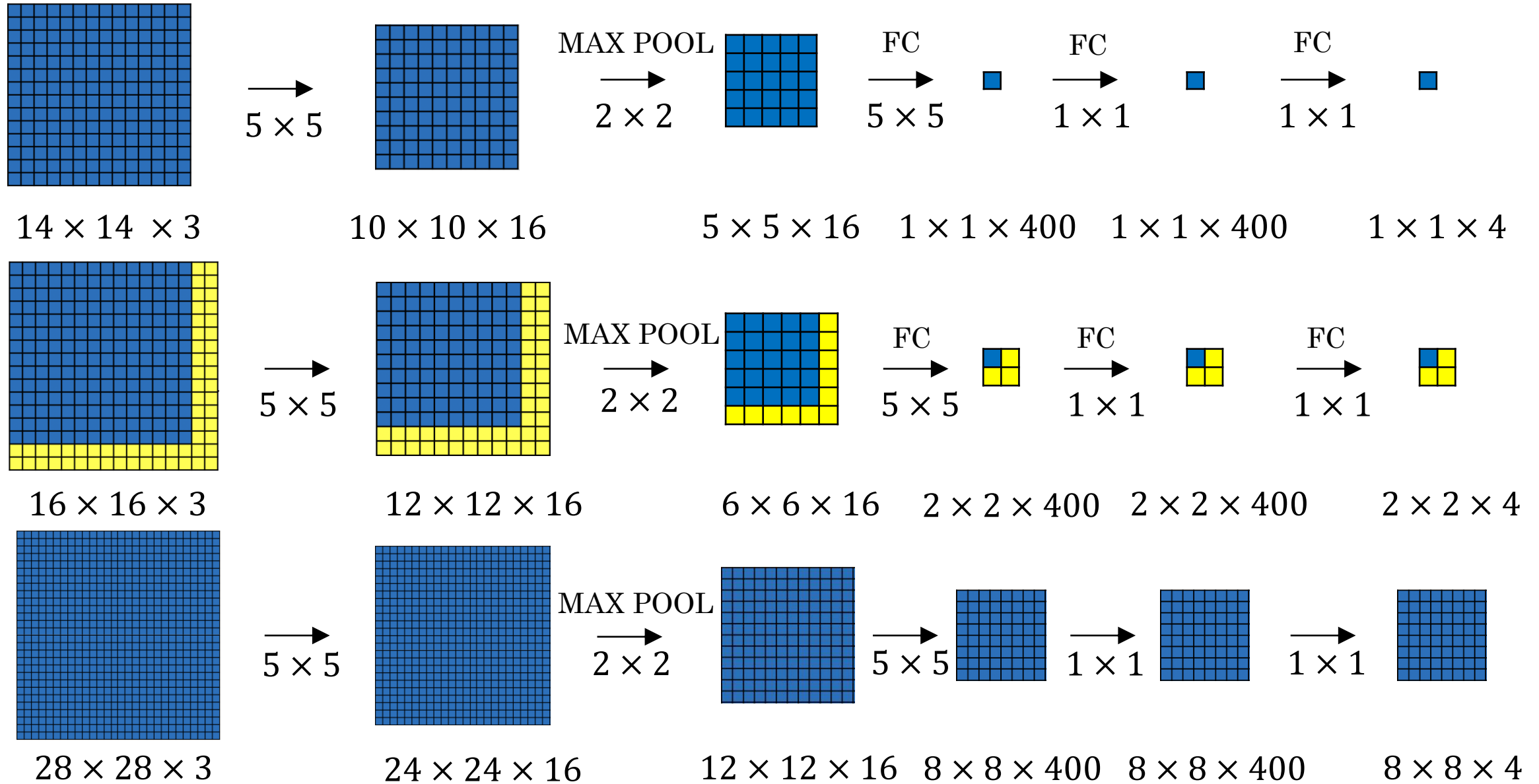
Object Detection

Convolutional
implementation of
sliding windows

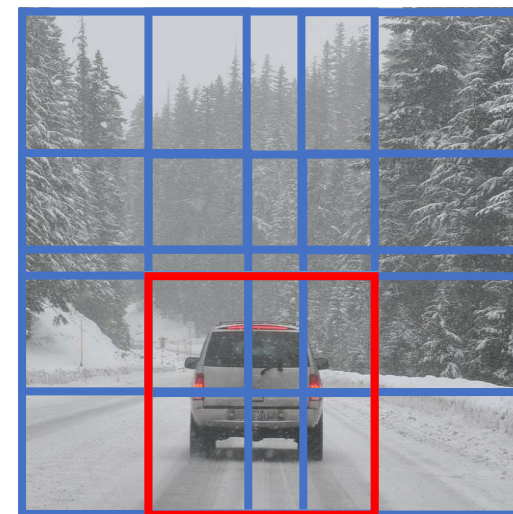
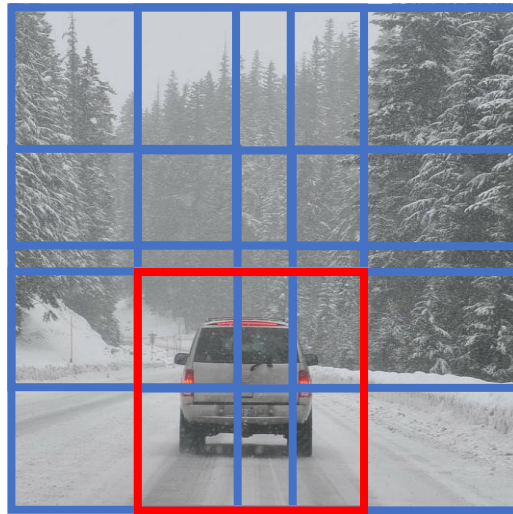
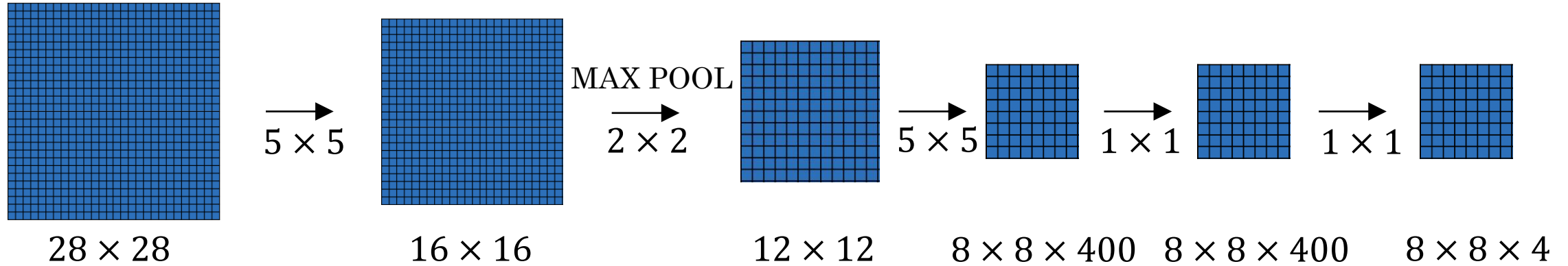
Turning FC layer into convolutional layers



Convolution implementation of sliding windows



Convolution implementation of sliding windows



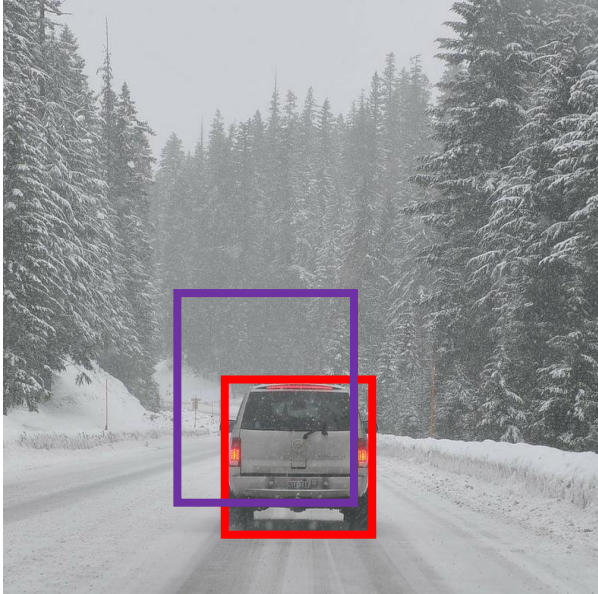


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Object Detection

Intersection
over union

Evaluating object localization



“Correct” if $\text{IoU} \geq 0.5$

More generally, IoU is a measure of the overlap between two bounding boxes.



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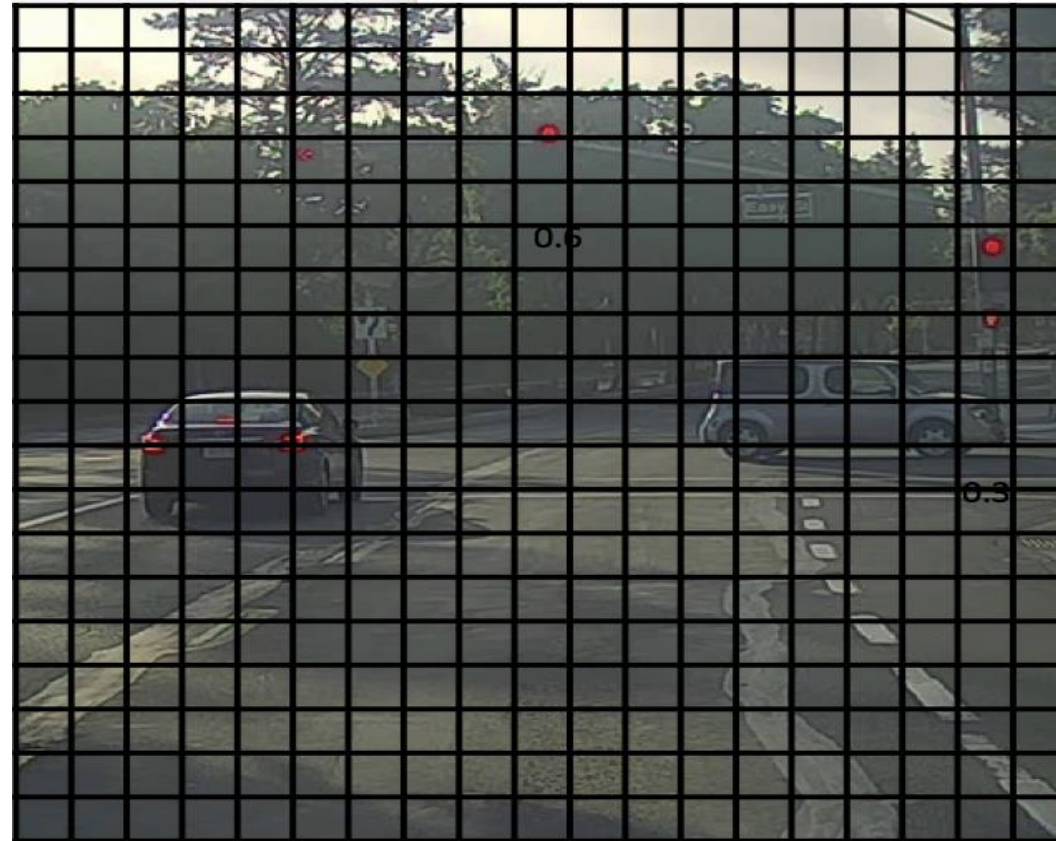
Object Detection

Non-max
suppression

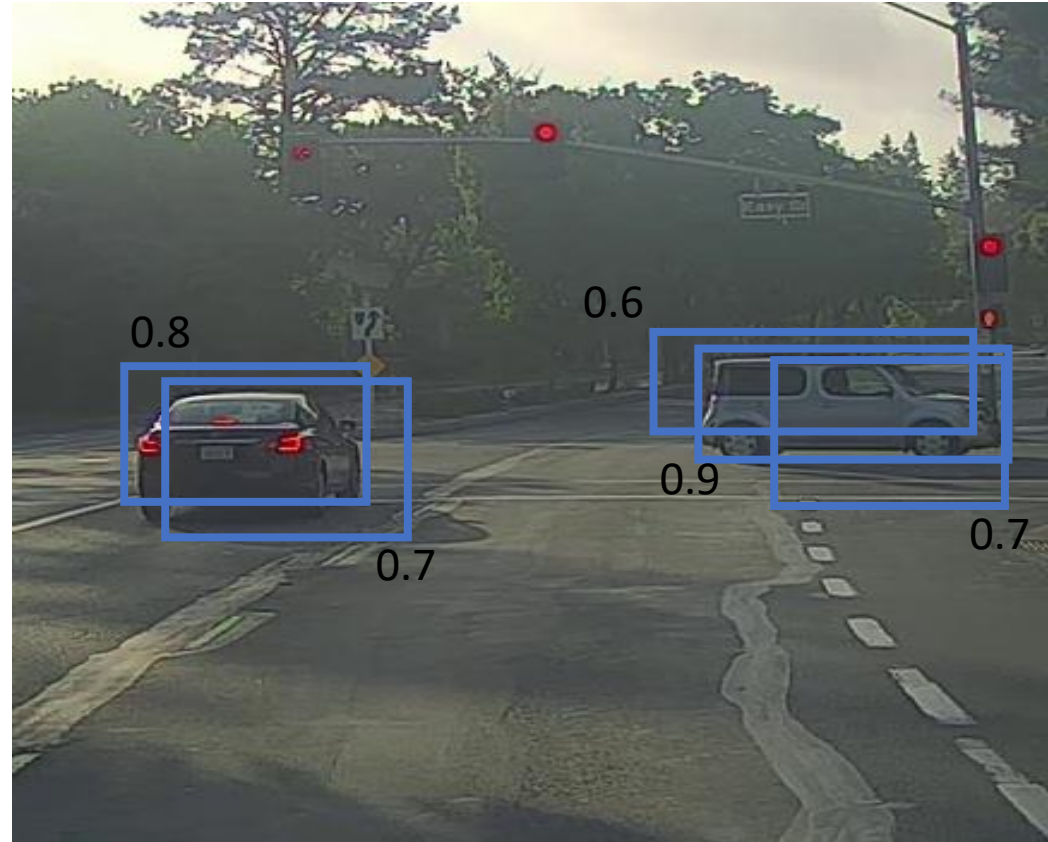
Non-max suppression example



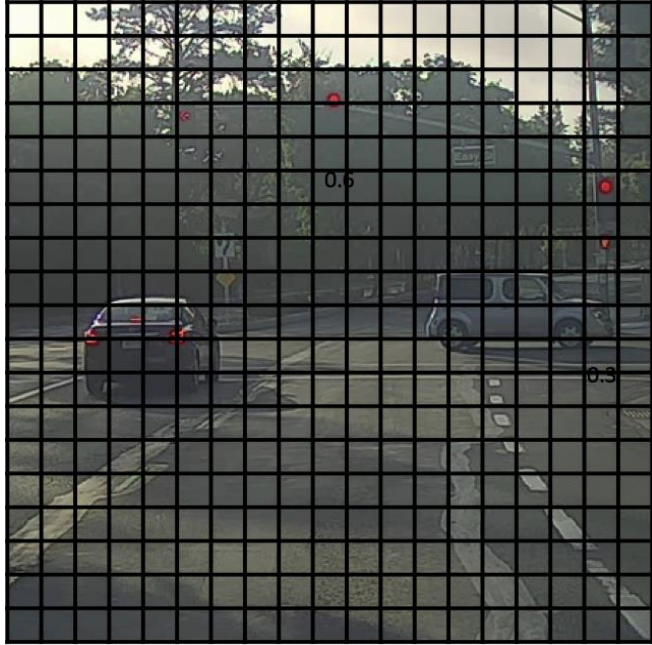
Non-max suppression example



Non-max suppression example



Non-max suppression algorithm



19× 19

Each output prediction is: $\begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \end{bmatrix}$

Discard all boxes with $p_c \leq 0.6$

While there are any remaining boxes:

- Pick the box with the largest p_c
Output that as a prediction.
- Discard any remaining box with $\text{IoU} \geq 0.5$ with the box output in the previous step

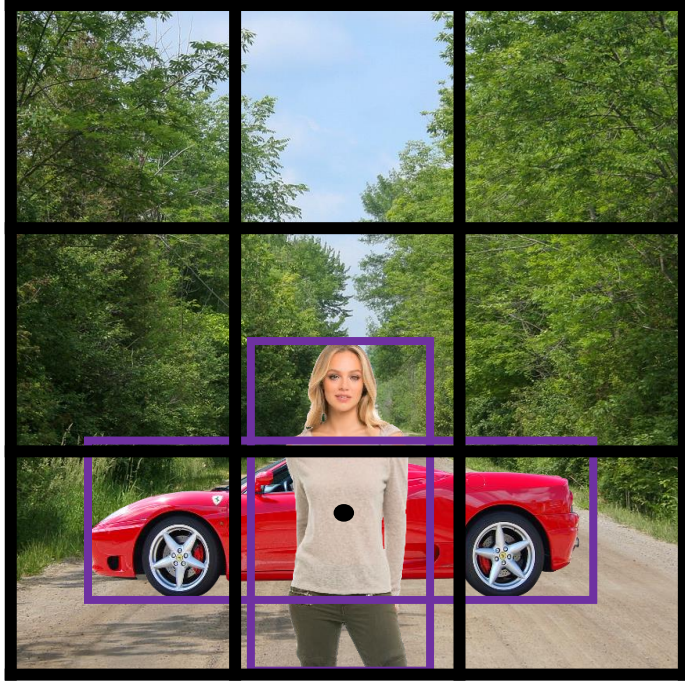


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Object Detection

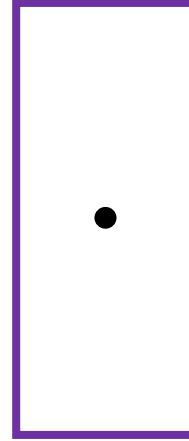
Anchor boxes

Overlapping objects:

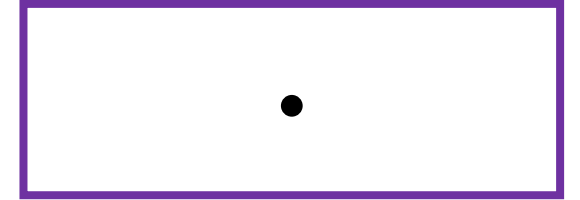


$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Anchor box 1:



Anchor box 2:



Anchor box algorithm

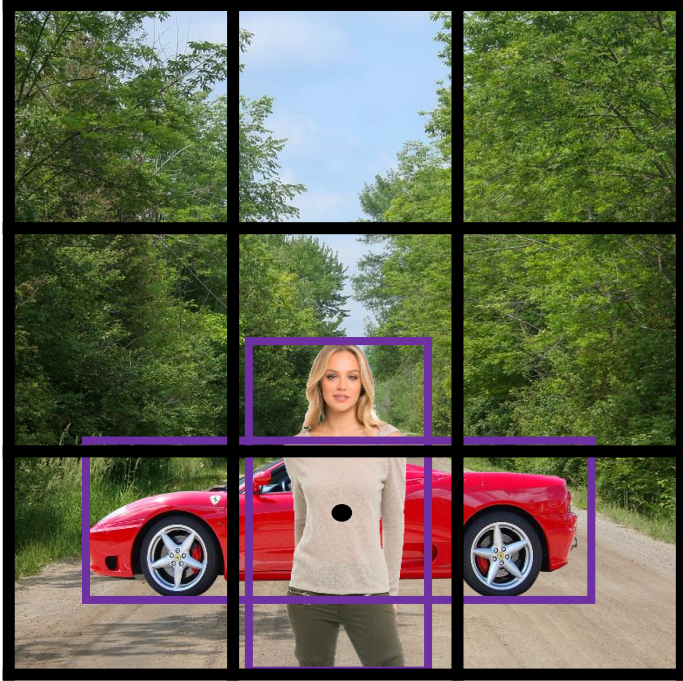
Previously:

Each object in training image is assigned to grid cell that contains that object's midpoint.

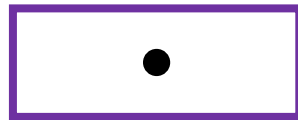
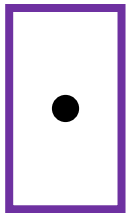
With two anchor boxes:

Each object in training image is assigned to grid cell that contains object's midpoint and anchor box for the grid cell with highest IoU.

Anchor box example



Anchor box 1: Anchor box 2:



$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$



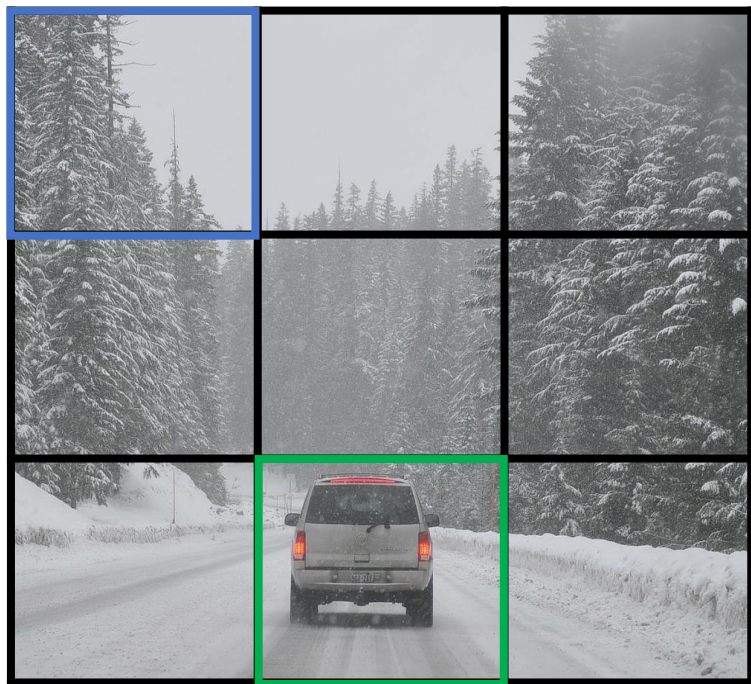
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Object Detection

Putting it together:
YOLO algorithm

Training

- 1 - pedestrian
- 2 - car
- 3 - motorcycle



$y =$

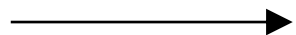
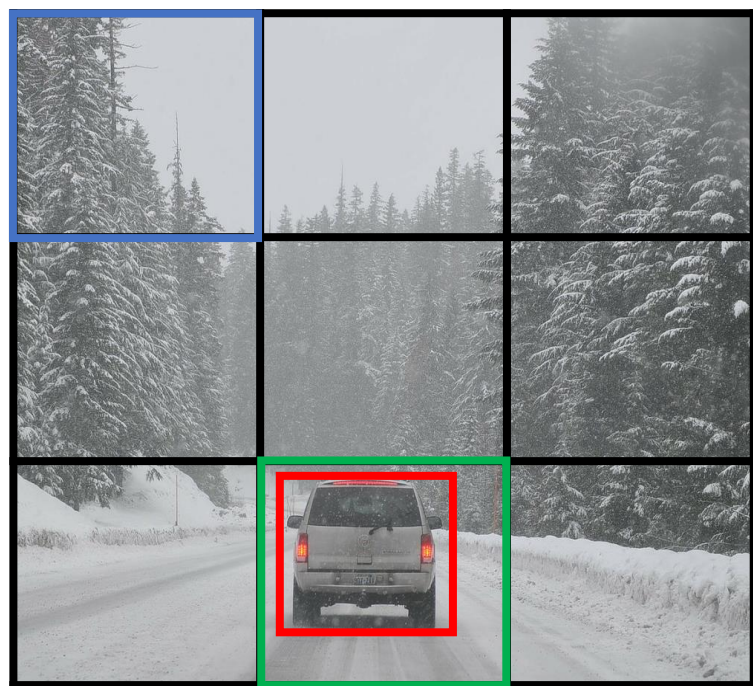
$$\begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix}$$

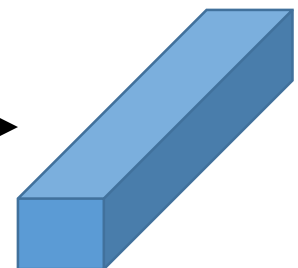
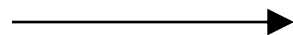
$$\begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

y is $3 \times 3 \times 2 \times 8$

Making predictions



...

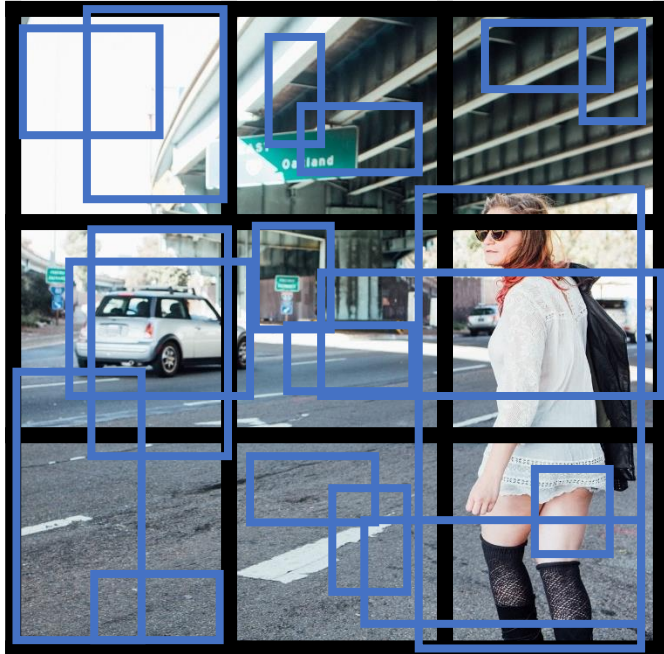


$3 \times 3 \times 2 \times 8$

$y =$

$$\begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Outputting the non-max suppressed outputs



- For each grid cell, get 2 predicted bounding boxes.
- Get rid of low probability predictions.
- For each class (pedestrian, car, motorcycle) use non-max suppression to generate final predictions.

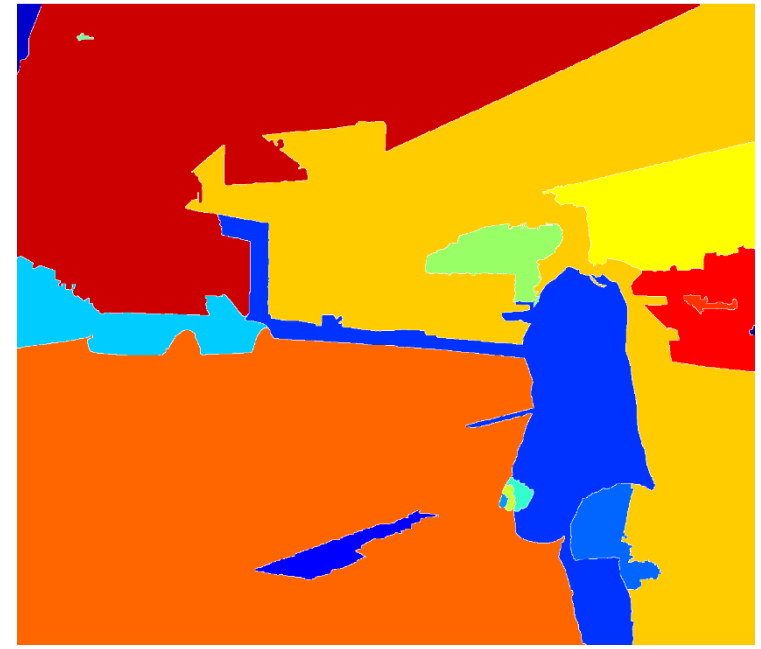
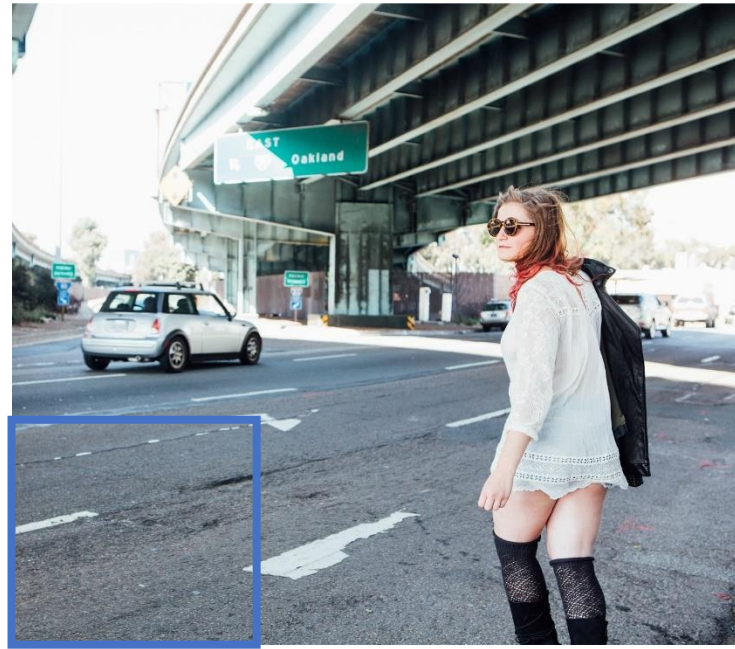
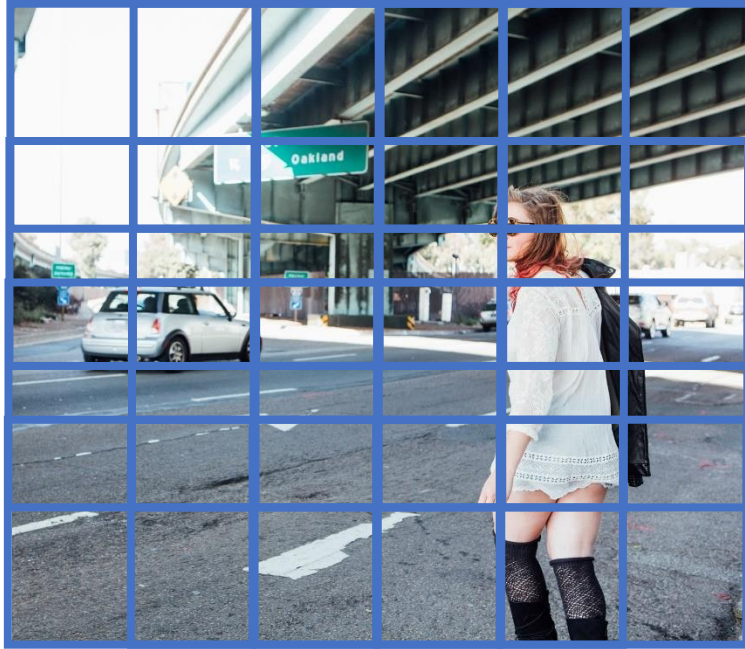


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Object Detection

Region proposals
(Optional)

Region proposal: R-CNN



Faster algorithms

R-CNN: Propose regions. Classify proposed regions one at a time. Output label + bounding box.

Fast R-CNN: Propose regions. Use convolution implementation of sliding windows to classify all the proposed regions.

Faster R-CNN: Use convolutional network to propose regions.

[Girshik et. al, 2013. Rich feature hierarchies for accurate object detection and semantic segmentation]

[Girshik, 2015. Fast R-CNN]

[Ren et. al, 2016. Faster R-CNN: Towards real-time object detection with region proposal networks]